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ITC 413: NoSQL Databases

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Assignment 11

Current State Analysis

TechFlow's current solution of a single MySQL relational database for its e-commerce platform, customer analytics dashboard and internal operations creates performance and scalability limitations because the three systems have very different workload requirements. Relational databases are strong for structured transactions but they become increasingly harder to scale when the applications also require high read/write volume, low latency response times and high availability (Sullivan, 2015). In TechFlow's case, all three systems are competing for database resources, which creates contention in the CPU, memory, storage and query executions. The e-commerce platform is rather difficult to manage in a single relational database because product catalogs often contain items with different attributes. In addition, some applications may require more flexibility in their model for the products with varying fields (Sullivan, 2015). Also, the customer analytics dashboard creates additional challenges because of the recommendation engines, user behavior tracking and sentiment analysis which all rely on high volume relationship heavy data. Overall, the single database model reduces the flexibility and limits TechFlow's ability to scale each system according to its own needs (Wu et al., 2018).

Database Selection Matrix

The database selection should be based on workload factors such as data volume, consistency needs, query patterns and schema flexibility rather than forcing one model to support every application (Sullivan, 2015). Below is the recommendation matrix for TechFlow.

System	Best Fit	Main Reason
E-commerce platform	Document database	Product data has varying attributes, so schema flexibility matters more than rigid relational structure; orders can still use relational support for strict transactions (Sullivan, 2015).
Customer analytics dashboard	Column family and graph	Analytics involves large and growing data volume plus recommendation and sentiment relationships; column family databases scale for heavy data workloads, while graph databases fit relationship-driven queries (Sullivan, 2015).
Internal operations	Relational database	Employee, project, and financial data require structured schemas, predictable queries, and strong consistency (Sullivan, 2015).

This matrix supports a polyglot persistence strategy with having document databases handle e-commerce, column family and graph databases for analytics and relational databases for internal operations (Sullivan, 2015).

Polyglot Architecture Proposal

TechFlow Solutions needs to move from a single MySQL database to a polyglot persistence architecture in which each major system uses the database type that best fits its workload. NoSQL and relational databases are complementary, so TechFlow will be able to choose the right tool for each job instead of forcing one database to support every workload. For the e-commerce platform, a document database is recommended to handle the product catalog

and to have a key-value database for shopping carts and session data. Having a document database is the best fit for the catalog because product records often have different attributes and these databases provide the schema flexibility needed for that type of data (Sullivan, 2015). A key-value store is appropriate here for carts and sessions because it will support simple, fast reads and writes for temporary web application data. The customer analytics dashboard should use a column family database for the large-scale event data and a graph database for recommendation logic and relationship-based analytics. Cassandra would be a strong fit as it supports time-series or event-driven data, large datasets that continue to grow and workloads built around query patterns (Wu et al., 2018). With internal operations, which include the employee directory, project management and financial reporting, a relational database is recommended. Relational systems remain strong for transaction processing because they protect data integrity and reduce anomalies, which makes them an ideal choice for internal business data. For data integration rather than having direct cross database joins, the use of APIs and event-driven synchronization with each database serving a system of record for its own domain. With shared identifiers such as customer ID, product ID and order ID to allow data to be connected across the systems. To support migration from MySQL successfully, a phased approach should be used rather than a full transfer. Product catalog data can be moved first to a document database; carts and sessions can then be moved to key-value storage. The analytic workloads can move over to column family and graph systems and then the internal operations can remain in MySQL until the final stage. Also, it is important to note that if NoSQL databases are utilized they require careful data modeling, so pilot deployments are also recommended before broader production use.

Business Impact Assessment

The recommended polyglot architecture would provide for a stronger long term business value than maintaining the current single MySQL database because it will align each system with the database model that is best suited for its workload. NoSQL databases were developed to address scalability, flexibility, cost and availability challenges that traditional relational systems struggle with at a larger scale (Sullivan, 2015). With TechFlow, this means the e-commerce platform would handle changing product attributes more efficiently, the analytics system would scale better for high volume and relationship driven workloads and internal operations would remain on a relational platform to ensure consistency and structured reporting. Another benefit of adopting this system is better performance isolation. Under the current single database approach, all three systems are competing for the same resources. So, when heavy analytics or catalog activity is happening it will reduce performance for order processing or internal reporting. With the polyglot design it will reduce the bottleneck by separating the workloads across systems better designed for those specific access patterns (Sullivan, 2015). However, with this comes the main risk which is increased complexity. Managing multiple databases will require more administration, more integration planning and more specialized skills. When these databases are used incorrectly or without sufficient experience it can create problems. Overall, the current single database is simpler to manage but comes at the risk of decreased performance over time when traffic becomes heavier. While the polyglot architecture offers better scalability, flexibility and workload separation, all things that are vital for the growth of TechFlow. And if used with a phased migration approach, the long-term benefits will outweigh the added complexity.

References

Sullivan, D. (2015). *NoSQL for Mere Mortals*. Addison-Wesley Professional.

Wu, X. (Brian), Kadamby, S., Kandhare, D., & Ploetz, A. (2018). *Seven NoSQL Databases in a Week*. Packt Publishing